Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			moles = 0.05 × 0.25 = 0.0125 (1) mass = 0.0125 × 106 = 1.325 g (1) dissolve mass in small volume of (deionised) water (1) add solution to a 250 cm^3 volumetric / graduated flask (1) using a funnel / wash beaker (1) make up to the mark and invert / shake well to mix (1)	4	2		6	1	6
	(b)	(i)		26.20 cm^3 - do not accept 26.2			1	1		1
		(ii)		$\frac{0.10}{26.30} \times 100 = 0.38$ %		1		1	1	1
		(iii)		e.g. funnel kept in burette (1) therefore value of titre less, since more acid dropped into burette from funnel (1) or difficult to see when indicator changed colour (1) therefore value of titre more, since end point overshot (1) or jet not filled / air bubble in burette (1) therefore value of titre more, since acid used to fill jet / bubble (1) or burette not rinsed with acid beforehand (1) therefore value of titre is more, since acid solution is more dilute (1)			2	2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		moles HCl = 0.090 × 0.02265 = 2.04×10^{-3} (1) moles NaOH unused in 25 cm^3 sample = 2.04×10^{-3} moles NaOH in volumetric flask = 2.04×10^{-2} (1) initial moles NaOH = 1.00 × 0.050 = 5×10^{-2} moles NaOH reacted = $5.00 \times 10^{-2} - 2.04 \times 10^{-2} = 2.96 \times 10^{-2}$ (1) allow ecf		2	1	3	2	3
		(ii)		moles $(NH_4)_2SO_4$ used = 1.48×10^{-2} mass $(NH_4)_2SO_4$ = $1.48 \times 10^{-2} \times 132.18 = 1.96$ g (1) percentage $(NH_4)_2SO_4$ = $\frac{1.96}{4.24} \times 100 = 46.2$ % (1) allow ecf from part (i) and for any calculated mass		2		2	1	
				Question 7 total	**4**	**7**	**4**	**15**	**5**	**13**